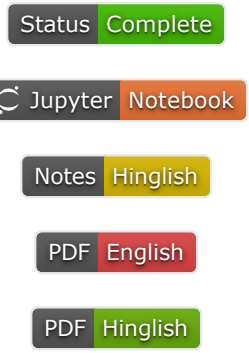
Lecture 30.08: XGBoost Code & NovaGen Clinical Trial Project



📌 Topic Overview

Practical implementation of XGBClassifier from the xgboost library. Covers data preparation, missing value handling, hyperparameter tuning, feature importance extraction, and classification evaluation.

📋 Key Learning Objectives

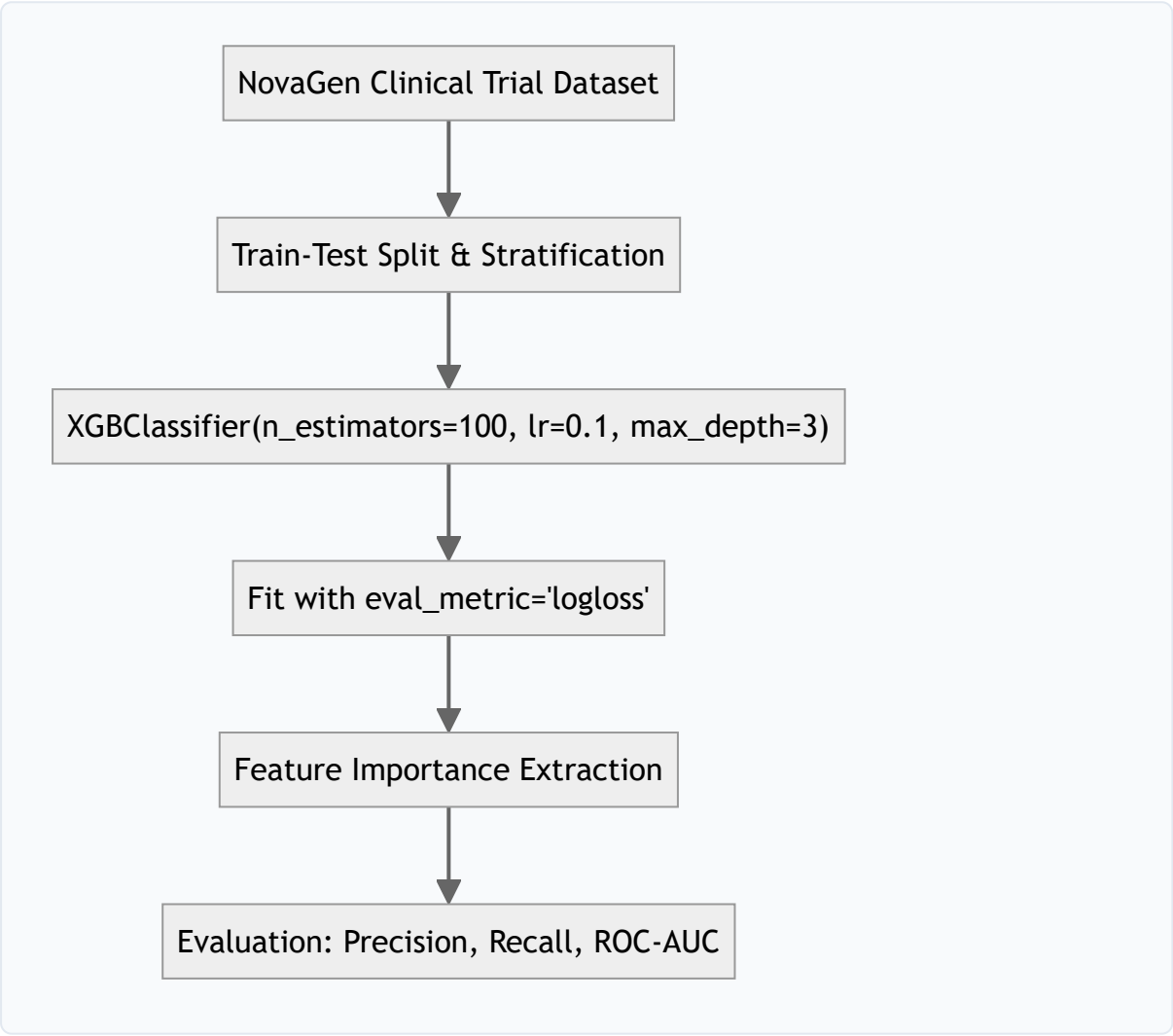
- Installing and importing xgboost.XGBClassifier.
- Configuring hyperparameters: n_estimators, learning_rate, max_depth, eval_metric='logloss'.
- Handling clinical trial patient features in the NovaGen dataset.
- Extracting and interpreting feature importances.
- Evaluating model performance with classification report and ROC-AUC.

🔍 In-Depth Conceptual Breakdown

- **XGBoost Installation & Import**: `from xgboost import XGBClassifier` ko import karke modern gradient boosting model banaya gaya.
- **Data Preprocessing**: NovaGen clinical trial dataset mein numerical aur categorical patient measurements ko encode kiya gaya.
- **Hyperparameter Setup**: `XGBClassifier(n_estimators=100, learning_rate=0.1, max_depth=3, eval_metric='logloss', random_state=42)` instantiate kiya gaya.

- **Model Training**: `xgb.fit(X_train, y_train)` se 2nd-order Taylor expansion gradients compute karke trees fit kiye gaye.
- **Feature Importances**: `xgb.feature_importances_` se reveal hua ki clinical biomarkers target class separation mein kitna bada role play karte hain.
- **Generalization**: Test set par evaluate karne par XGBoost ne baseline decision trees aur Random Forest se superior performance deliver ki.

📊 Architectural Flowchart



🔢 Mathematical Formulation

Taylor expansion of loss function around previous prediction $\hat{y}^{(t-1)}$:

$$\mathcal{L}^{(t)} \approx \sum_{i=1}^{N} \left[l(y_i, \hat{y}^{(t-1)}) + g_i f_t(x_i) + \frac{1}{2} h_i f_t^2(x_i) \right] + \Omega(f_t)$$

Optimal leaf weight w_j^* for leaf j containing sample set I_j:

$$w_j^* = -\frac{\sum_{i \in I_j} g_i}{\sum_{i \in I_j} h_i + \lambda}$$

💻 Implementation

The fully tested and verified standalone implementation is provided in the synchronized Jupyter Notebook.

🔬 Capstone Project Reference: NovaGen Research Labs

The complete end-to-end clinical trial classification project using XGBoost, Random Forest, and Voting Ensembles is located in: 👉 **NovaGen_Lab_Project/**

- **Notebook**: `NovaGen_Lab_Project/novagen.ipynb`
- **Dataset**: `NovaGen_Lab_Project/novagen_dataset.csv`
- **Problem Statement**: `NovaGen_Lab_Project/Supervised ML _Assignment5.pdf`